# Jack Basinski

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## Education

**University of Washington** — Seattle, WA
*Bachelor of Science in Electrical and Computer Engineering* — *September 2023 – Present*

- GPA: 3.6, Honors student, Dean's List scholar.
- **Relevant Coursework**: Applied Electromagnetics, Computer Architecture, Programming for Signal Processing, Semiconductor Device Physics, Design of Digital Circuits and Systems.

**University of Hawaii** — Oahu, HI
*College-level Mandarin Chinese coursework, StarTalk NSA initiative.* — *June 2022 – July 2022*

## Experience

**Undergraduate Research Assistant** — May 2025 – Present
*Washington Nanofabrication Facility* — *Seattle, WA*

- Execute semiconductor manufacturing processes to support 150+ academic and commercial facility users.
- Characterized photoresist spin curves, increasing production efficiency.
- Enforce ISO 5 cleanroom operational integrity.

**Lab Assistant** — September 2024 – Present
*UW Nanodevice Physics Group* — *Seattle, WA*

- Assemble nanodevices using VDW materials for research in energy, information, and sensing applications.
- Fabricated a 16-channel feedback lock-in amplifier for efficient multi-terminal transport measurements. Enabled parallelization of data collection, **saving $97,425 in lab expenses.**

**Checker/Customer Service Rep** — June 2023 – Present
*Metropolitan Market Grocery* — *Seattle, WA*

- Support and assist customers to maintain excellent guest experience at upscale market.

## Projects

**Schottky Diode Fabrication** | *Deposition, Etch, Lithography, Metrology* — June 2025 – July 2025

- Utilized DELM processes to pattern features ranging 5-100µm on 4-inch wafers.
- Characterized diode behavior with 4-terminal impedence measurements, film thickness verification, IV performance. **Achieved ideality factors within 20% of theoretical predictions.**

**Pipelined CPU** | *SystemVerilog, Assembly, Modelsim* — June 2025 – August 2025

- Designed a 5-stage pipelined CPU around ARM's LEGV8 instruction set.
- Simulated and validated performance across benchmark algorithms, including quicksort and fibonacci, demonstrating **x2.7 speedup over non-pipelined implementation.**
- Implemented forwarding hardware solutions, eliminating data hazards and **improving CPI by 70%**

**Autonomous Line-Following Robot** | *C, Soldering, PCB Design* — September 2023 – December 2023

- Designed a PCB and Arduino-based control system, enabling precision sensor integration.
- Implemented PID control to optimize path tracking, **achieving sub-60 second course completion.**

**Misc Summer Projects** | *C, Soldering, Mig Welding, Shop tools* — June 2020 – Present

- Designed and built custom RC boat using Arduino and nRF24L01 RF modules.
- Retrofitted cruiser bicycle with a 100cc two-stroke engine.
- Designed, fabricated, and field-tested a motorized all-terrain go-kart. Experimentally improved original design.

## Technical Skills

**Languages**: SystemVerilog, Assembly, Java, Python, C, MATLAB.
**Engineering Tools**: Nanofabrication Process Tools (DELM), FPGA Development, Modelsim, PCB Fabrication, Soldering, SolidWorks, Microsoft Excel, Shop Tools, MIG Welding.
**Libraries**: pandas, NumPy, Matplotlib.
**General Knowledge + Interests**: Mandarin Chinese, Photography, Hiking, Stevie Ray Vaughn.